JAMES DE BROECK

Software Development Engineer

CONTACT

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SKILLS

Object-Oriented Programming 5 years +

Parallelized Programming 5 years +

Code Optimization

TECHNOLOGIES

C 5 years +

Python 4 years +

C++

CI/CD

WORK EXPERIENCES

Software Development Engineer II — Feb 2018 - Present

Amazon Web Services (AWS)

- Collaborated on development of AWS Elastic Fabric Adapter (EFA), which delivers low latency networking for scalable high performance computing (HPC). Worked on operational APIs, where critical thinking was required to create efficient and scalable, but easily usable interfaces.
- Increased efficiency of firmware-based metrics collection a thousand-fold by compressing and caching data used in API calls, resulting in higher resolution metric data used to make critical business decisions.
- Lead the development of a parallel-based behavioral driven development (BDD) testing framework, scaling to thousands of real-world AWS cloud-based testing scenarios in order to continuously validate and safely release new software to AWS infrastructure world-wide.
- Grew team's CI/CD infrastructure to further automate release testing in order to reduce time between patch creation and deployment, therefore resolving customer issues sooner.

PROJECTS

SwarmEngine - Capstone Project — 2017

Pacific Lutheran University

github.com/debroejm/SwarmEngine

- Project was a parallelization focused game engine, written from scratch in C++.
- Worked on the project alone, with a primary focus on rendering in separate thread/s from main loop logic.
- OpenCL was also explored to run computationally heavy but parallelizable tasks on the GPU.

EDUCATION

Bachelor of Science (B.S.), Computer Science and Computer Engineering — 2014 - 2018

Pacific Lutheran University